

Atharva Lele

Pune, India

9359031008 — athuvlele01@gmail.com — LinkedIn — GitHub

IIoT Engineer — Edge Systems — Backend Engineer

SUMMARY

IIoT and Backend Engineer with experience designing and deploying industrial edge systems across 10+ factories for real-time machine data ingestion, monitoring, and predictive maintenance. Strong expertise in industrial communication protocols, distributed systems, edge-to-cloud architectures, Docker, Kubernetes, and scalable backend development.

EXPERIENCE

Wipro – Associate Product Development Engineer (IIoT) Mar 2024 – Present

- Developed and deployed an edge-side IIoT platform across 10+ factories enabling real-time machine data collection and monitoring.
- Integrated industrial protocols including OPC-UA, Modbus, Profibus, HART, and DLMS for reliable telemetry acquisition from industrial assets.
- Designed containerized microservices using Docker and Kubernetes for protocol services, messaging infrastructure, and edge applications.
- Built a reusable protocol SDK reducing onboarding effort for new industrial devices and integrations.
- Developed a secure credential management module supporting encrypted storage and authentication workflows.
- Resolved 10+ production incidents in live deployments, improving reliability and reducing operational downtime.

PERSONAL PROJECTS

ReAct Agent (LangChain)

- Built an autonomous AI agent capable of multi-step reasoning and web-based market research using the ReAct architecture.

Multi-Agent System (AutoGen)

- Developed a collaborative AI system where specialized agents perform planning, research, and financial analysis workflows.

RAG Document Intelligence System

- Built a retrieval-augmented question-answering system over private document repositories using vector search and LLMs.

PROJECTS

Smart Street Litter Detection System

- Developed a computer vision pipeline using HOG-based human detection and YOLO-based object detection to identify littering events and automatically log violations.

Dual-Axis Solar Tracking System

- Designed and implemented an automated solar tracking system that dynamically adjusts panel orientation to improve energy capture efficiency.

EDUCATION

Bachelor of Engineering (Electronics & Telecommunication)

Pimpri Chinchwad College of Engineering, Pune

2019 – 2023

CGPA: 8.76

SKILLS

Languages: Python, SQL

Backend: FastAPI (Intermediate)

IIoT Protocols: OPC-UA, Modbus, Profibus, HART, DLMS

DevOps: Docker, Kubernetes, Linux

AI/LLM: LangChain, AutoGen, RAG, LLM APIs

Messaging: MQTT, EMQX, etcd

Cloud: AWS